[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
BORING MACHINES, ENGINE CYLINDER (PORTABLE)

The General Services Administration has authorized the use of this Commercial Item Description, for all Federal Agencies.

1. SCOPE. This description covers two sizes of portable boring machines for reboring engine cylinders. These machines are intended for use in base and field automotive repair shops.

2. CLASSIFICATION. The boring machines covered by this description shall be of the following sizes as specified herein:

Size 1 - Cylinder diameters 2.60 through 5.25 inches and a bore stroke of 12.50 Inches.

Size 2 - Cylinder diameters 3.50 through 7.50 inches and a bore stroke of 18 inches.

3. SALIENT CHARACTERISTICS.

3.1 Mounting devices. Mounting devices shall be provided to anchor the boring machines securely to a cylinder block and shall incorporate means for accurate positioning to the mean center of the cylinder to be bored. The bar shall be positioned within 0.002 of an inch of the true center of the cylinder in the unworn area of the cylinder.

3.2 Lifting provisions. Where suitable boring machine surfaces are not available for shop hoist handling, a lifting eye or hook shall be provided. The eye or hook shall have a lifting capacity of three times the weight of the boring machine. Should a lifting eye be used, the eye shall have an inside diameter of not less than two inches. If an open hook is used, to lift the machine, it shall have an inside diameter of no less than one inch.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document, should be sent by letter to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSTA-AR-EST-P Rock Island, IL 61299-7300.

FSC 4910

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3.3 Capacity.

3.3.1 Size 1. The Size 1 boring machine shall be capable of boring cylinder diameters in a size range of from at least 2.60 to no less than 5.25 inches, with a bore stroke of at least 12.50 inches.

3.3.2 Size 2. The Size 2 boring machine shall be capable of boring cylinder diameters in a size range of from at least 3.50 to no less than 7.50 inches, with a bore stroke of at least 18 inches.

3.3.3 Depth of cut. The boring machine shall be capable of producing a rough-cut depth of 0.040 inch and a finish-cut depth of 0.015 inch in a cast iron engine block. Each depth of cut specified above shall be accomplished in a single pass.

3.4 Accuracy. When the boring machine is operated at the finish-cut depth as stated in paragraph 3.3.3 above, the finished bore shall be within the following limits:

Diameter:	+ or -0.001 inch;
Taper:	0.001 inch in maximum length bore;
Out-of-roundness:	0.001 inch;
Surface Roughness:	63 micro inches, arithmetical average.

3.5 Motors. The motor furnished shall be rated at not less than ½ horsepower, continuous duty rating. The motor shall be the induction type, operating on 115 volt, 60 hertz, single-phase power, conforming to the requirements of NEMA MG-1.

3.6 Power cord. Each boring machine shall be furnished with not less than eight feet of three-wire No. 14 AWG, NFPA 70, type SO heavy duty, power cord. The cord shall be provided with a three-prong grounding type connector plug of either the grip cord type or molded on type.

3.7 Accessories. The following accessory equipment shall be provided with each boring machine furnished:

3.7.1 Tool holder. The tool holder mounted on the spindle head shall provide quick and easy micrometer adjustment of the cutting tool(s) to any cutting diameter within the capacity of the boring machine.

3.7.2 Tool sharpener. Tool sharpening facilities shall be incorporated as a part of the machine and shall be capable of producing a cutting edge of the proper form, rake, and clearance.

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3.7.3 Carrying case. A carrying case, made of wood or steel, shall be furnished with each boring machine. The case shall contain, protect, and carry the boring machine and its component parts, equipment, tools, and spare parts. The case shall be fitted with a hinged lid embodying positive

acting latches. The case shall have a carrying handle or handles suitable for man handling or sling handling.

3.7.4 Carbide tools, two complete sets.

3.7.5 Micrometer, with the required tools and attachments for measuring, calibrating, and positioning the cutter tools for boring and sharpening operations.

3.7.6 Lapping compound and lap impregnate.

3.8 Protective finish. The protective finish shall be in accordance with the best practice of the commercial industry.

3.9 Product identification.

3.9.1 Plates. Instruction and identification plates shall be made of a corrosion resistant material and be securely attached to the equipment. All wording shall be in the English language, using plain boldface letters. Lettering shall be permanently marked and have a contrasting background.

3.9.2 For Military agencies. The identification plate shall contain at least the following data:

Nomenclature	Manufacturer's serial number
National Stock Number (NSN)	Contract number
Manufacturer's name	Date of manufacture.
Manufacturer's model number	

3.9.3 For Civil agencies. The identification plate shall contain the name or trademark of the manufacturer, along with the capacity of the machine.

4. REGULATORY REQUIREMENTS.

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor conformance.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability. The manufacturers of the boring machine shall certify that the offered machine meets the following Market Acceptability Requirements:

5.2.1 All gears, pinions, and other cutter head drive system components operate for five years without failure or wear, which cannot be compensated by adjustments, or which would cause the boring machine to lose its inherent accuracy, provided the machine is operated and maintained in accordance with the manufacturer's operational instruction manual.

5.2.2 Also all drive system components, including tubes, drive shafts, columns, bushings, bearings, plates, etc., shall operate with out replacement for five years after delivery of the machines.

5.2.3 The boring machine is a market tested and commercially accepted item by the U.S. automotive rebuild industry, with a sales rate of at least 25 items per year or 100 items over the last 10 years.

6. PACKAGING

The packaging, packing, preservation, labeling, and marking shall be as specified in the contract or order.

7. NOTES.

The following specifications and standards forma part of this document to the extent specified herein:

a. NEMA MG 1 - Motors and Generators
b. NFPA 70 - National Electric Code

NFPA specifications are available from the National Fire Protection Association, one Batterymarch Park, PO Box 9101, Quincy, MA 02269-9101.

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NEMA standards are available from the National Electrical Manufacturers Association, 2101 L Street N. W., Washington, DC 20037.

MILITARY INTERESTS:

Custodians
Army - AL
Navy - YD1
Air Force -99

Review Activities
Army - ER, SM
Navy - MC, SH
Air Force -82

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 4910-0817

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